

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	K Madan Mohan Reddy	Reg. No.:	192411206
Date:	01-08-2026	Duration:	60 minutes

Code of Conduct

I K Madan Mohan Reddy (192411206) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Byreddy _____

Instructions to Students: "Online Loan Eligibility Prediction and Approval Management Platform"

1. Identify the Project Vision and Stakeholders

Define the project's vision, objectives, and identify all key stakeholders involved.

The Online Loan Eligibility Prediction and Approval Management Platform is designed to provide a secure, efficient, and intelligent solution for automating the loan application and approval process. The vision of the project is to help banks and financial institutions make faster, more accurate, and transparent lending decisions by using machine learning to predict an applicant's loan eligibility. The main objectives of the project are to simplify online loan applications, reduce manual verification, improve the accuracy and consistency of approval decisions, minimize processing time, and ensure secure storage and management of customer information. The platform also aims to enhance customer satisfaction by providing quick updates on application status and enabling administrators to efficiently manage loan requests.

- The key stakeholders involved in the project include loan applicants, who submit applications and track their status.
- Loan officers, who verify customer details and review eligibility predictions; bank managers or administrators, who make final approval or rejection decisions and monitor the overall process; system administrators, who maintain the platform and ensure data security.
- Software developers and testing teams, who design, develop, test, and maintain the application; database administrators, who manage the storage and integrity of customer data; and regulatory authorities, who ensure that the platform complies with banking regulations and data privacy standards.
- Together, these stakeholders contribute to delivering a reliable, efficient, and user-friendly loan management system.

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2. Create Epics and User Stories

Break down the requirements into Epics and formulate well-structured User Stories using the format:

As a <user>, I want <feature>, so that <benefit>.

User Registration and Authentication

- User Story 1: As a loan applicant, I want to register an account, so that I can apply for loans online.
- User Story 2: As a registered user, I want to log in securely, so that I can access my loan application details.
- User Story 3: As a user, I want to reset my password, so that I can regain access to my account if I forget it.

Loan Application Management

- User Story 1: As a loan applicant, I want to fill out and submit an online loan application, so that I can request a loan without visiting the bank.
- User Story 2: As a loan applicant, I want to upload the required documents, so that my application can be verified quickly.
- User Story 3: As a loan applicant, I want to edit my application before submission, so that I can correct any mistakes.

Loan Eligibility Prediction

- User Story 1: As a loan applicant, I want to receive an instant eligibility prediction, so that I know my chances of loan approval.
- User Story 2: As a loan officer, I want to view the machine learning prediction, so that I can make informed approval decisions.

Loan Approval Process

- User Story 1: As a loan officer, I want to review loan applications, so that I can verify applicant information.
- User Story 2: As a bank manager, I want to approve or reject loan applications, so that eligible customers receive loans promptly.
- User Story 3: As a loan applicant, I want to receive notifications about my application status, so that I stay informed throughout the process.

Administration and Reports

- User Story 1: As an administrator, I want to manage user accounts, so that I can maintain system security and access control.
- User Story 2: As an administrator, I want to generate loan reports, so that I can monitor application and approval statistics.
- User Story 3: As an administrator, I want to view all loan applications, so that I can oversee the entire loan management process.

3. Define Acceptance Criteria

- **Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.**

The acceptance criteria for the Online Loan Eligibility Prediction and Approval Management Platform ensure that each user story can be tested and verified before deployment. For the user registration feature, given that a new user enters valid personal details, when they submit the registration form, then the system should successfully create an account and display a confirmation message. For secure login, given that a registered user provides valid credentials, when they click the login button, then they should be granted access to their dashboard; otherwise, an appropriate error message should be displayed. For the loan application feature,

given that the applicant is logged in, when all mandatory details and required documents are submitted, then the application should be stored successfully in the database. For loan eligibility prediction, given that a complete application has been submitted, when the machine learning model evaluates the applicant's information, then the system should display the eligibility result as either "Eligible" or "Not Eligible."

- ✓ For the loan approval process, given that a loan officer or bank manager reviews the application, when they approve or reject it, then the application status should be updated and the applicant should receive a notification.
- ✓ Finally, for administration and reporting, given that an administrator selects a report type and time period, when the report is generated, then the system should display accurate loan statistics and provide an option to download the report in a suitable format such as PDF or Excel.
- ✓ These acceptance criteria ensure that the platform functions correctly, meets user requirements, and delivers reliable and measurable outcomes.

4. Prioritize the Product Backlog

- **Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.**

The product backlog for the Online Loan Eligibility Prediction and Approval Management Platform is organized and prioritized based on business value, customer needs, and project dependencies to ensure that the most critical features are developed first. The highest priority is given to user registration and secure login, as these are essential for users to access the platform. The next priority is the online loan application module, which enables applicants to submit their loan requests and upload the required documents. Once the application process is established, the machine learning-based loan eligibility prediction feature is implemented to provide accurate and instant eligibility results. This is followed by the loan approval and rejection module, allowing loan officers and bank managers to review applications and make informed decisions.

- Notification services are then integrated to keep applicants updated on the status of their loan requests.
- Finally, administrative features such as user management, report generation, system monitoring, data backup, and security management are implemented to ensure smooth operation and maintenance of the platform.
- This prioritized backlog ensures that essential business functions are delivered first, customer expectations are met efficiently, and all dependent features are implemented in the proper sequence for the successful completion of the project.

5. Plan the Sprint Backlog

- **Select high-priority user stories for the first sprint and divide them into actionable development tasks.**

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6. Estimate Effort Using Story Points

- **Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.**

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7. Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

The sprint goal for the Online Loan Eligibility Prediction and Approval Management Platform is to develop and deliver the core functionalities that enable users to register, log in securely, submit loan applications, and receive an initial loan eligibility prediction. During this sprint, the development team will focus on implementing essential features that provide a functional foundation for the platform. The expected sprint deliverables include a user registration and authentication module, an online loan application form with document upload capability, integration of the machine learning model for loan eligibility prediction, and a dashboard that displays the prediction results to applicants. Basic validation, secure data storage, and database connectivity will also be completed to ensure reliable system performance. By the end of the sprint, users should be able to create an account, log in, submit a loan application, upload the required documents, and receive an eligibility prediction, providing a working prototype that supports the platform's primary business objectives.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation

	professional and well organized.		adequate but incomplete.	are unclear or incomplete.
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Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25